

Quiz 11

A few weeks ago, we were thinking about rain in a rainforest. We modeled the daily rainfall, measured in inches, using iid random variables X_i with $X_i \sim \text{exponential}\left(\frac{365}{80}\right)$.

1. What are $E[X_i]$ and $\text{Var}(X_i)$?

$$E[X_i] = \frac{1}{\frac{365}{80}} = \frac{80}{365}$$

$$\text{Var}(X_i) = \frac{1}{\left(\frac{365}{80}\right)^2} = \left(\frac{80}{365}\right)^2$$

2. Let Y be the total rainfall in the rainforest for a year (365 days), according to the model. Calculate $E[Y]$, $\text{Var}(Y)$, and $\sigma(Y)$, including units of measurement for $E[Y]$ and $\sigma(Y)$. Use the methods of Chapter 6 to explain your calculations of $E[Y]$ and $\text{Var}(Y)$.

$$E[Y] = 80 \text{ inches}$$

Explanation:

The total rainfall for the year is the sum of the rainfall for the 365 days in the year: $Y = X_1 + X_2 + \cdots + X_{365}$; therefore, $E[Y] = E[X_1] + E[X_2] + \cdots + E[X_{365}]$. The expected rainfall for each day is $80/365$, and so $E[Y] = 365 \left(\frac{80}{365}\right) = 80$.

$$\text{Var}(Y) = \frac{80^2}{365}$$

Explanation:

Since the daily rainfall amounts (X_i) are independent, $\text{Var}(Y) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_{365})$. The variance for each daily rainfall variable is $\left(\frac{80}{365}\right)^2$, and so $\text{Var}(Y) = 365 \left(\frac{80}{365}\right)^2 = \frac{80^2}{365}$.

$$\sigma(Y) = \frac{80}{\sqrt{365}} \text{ inches or about 4.2 inches}$$

3. Let $\bar{Y}$ be the random variable representing the average daily rainfall for an entire year, according to this model. Write a formula for $\bar{Y}$ in terms of the X_i .

$$\bar{Y} = \frac{1}{365}(X_1 + X_2 + \cdots + X_{365})$$

4. Calculate $E[\bar{Y}]$, $\text{Var}(\bar{Y})$, and $\sigma(\bar{Y})$, including units of measurement for the expected value and standard deviation.

$$E[\bar{Y}] = \frac{80}{365} \text{ inches per day or about 0.22 inches per day}$$

$$\text{Var}(\bar{Y}) = \frac{80^2}{365^3}$$

$$\sigma(\bar{Y}) = \frac{1}{\sqrt{365}} \left(\frac{80}{365}\right) \text{ inches per day or about 0.011 inches. That's a standard error of the mean—small!}$$